Non-Bloch Band Theory

Generalised Brillouin zone, non-Bloch winding number,
and how far the open spectrum sits from $E(\mathrm{GBZ})$

Kevin

Abstract

Bloch's theorem fails for lattices with a non-Hermitian skin effect: open-chain eigenstates grow or decay as β^n with $|\beta| \neq 1$, and the periodic spectrum is unrelated to the open one. Non-Bloch band theory replaces the Brillouin zone $|\beta| = 1$ by the generalised Brillouin zone (GBZ). We compute the GBZ with a single θ-scan root-finding method for three chains: Hatano–Nelson, the non-reciprocal SSH chain (as one band in $\lambda = E^2$), and a range-2 chain whose GBZ is not a circle. The scan reproduces the analytic circular GBZs to 10^{-15}. The non-Bloch winding number matches the open-chain edge-state count everywhere except within 0.072 of a transition, while the Bloch winding takes the non-integer value $1/2$ across most of the phase diagram. We derive, from a Laplace expansion of the boundary determinant, the leading finite-size term: each open eigenvalue misses the GBZ condition by exactly $\ln|C(E)|/(N+1)$, with C computed from the characteristic roots alone; this matches the numerics to a median of 10^{-12} with no fitted parameter. We give the exact condition for a circular GBZ, confirm it by a parameter scan, and show that an imaginary gauge transform keeps the eigensolver accurate to 10^{-13} at $N = 100$ where the plain solver errs by 2.7×10^{-7}.

1 Introduction

For a Hermitian lattice, periodic and open boundary conditions give the same bulk spectrum as $N \to \infty$, and topological invariants computed on the Brillouin zone (BZ) predict edge states. For non-Hermitian lattices with non-reciprocal hopping, all bulk eigenstates of the open chain pile up at one edge, and the open spectrum is a set of arcs lying *inside* the loop traced by the periodic spectrum [4, 3]. Yao and Wang [1] and Yokomizo and Murakami [2] showed that the correct object is the GBZ: the curve of complex β from which open-chain eigenstates are built. Mathematically this is the Schmidt–Spitzer theorem for banded Toeplitz matrices [8, 9].

This project builds one GBZ solver, validates it on chains with closed-form answers, connects it to the biorthogonal picture of the companion nh-SSH project, and goes beyond the $N \to \infty$ statement by deriving *how* the finite open spectrum approaches $E(\mathrm{GBZ})$.

Contributions.

1. A model-agnostic θ-scan GBZ solver with root-counting acceptance, applied to one- and (chiral) two-band chains.
2. Non-Bloch versus Bloch winding against the open-chain edge-state count, with the absolute-value form of the transition condition and the half-integer Bloch winding explained by zero counting.
3. A derived, parameter-free leading finite-size term $\ln|C(E)|/(N+1)$.
4. The exact circular-GBZ criterion, a deformation scan, and an imaginary-gauge fix for eigensolver accuracy.

Table 1: Summary of the main results.

Quantity	Computed	Analytic
Hatano–Nelson GBZ radius, three (t_R, t_L)	$\sqrt{t_R/t_L}$ to $\le 2\times10^{-15}$	$\sqrt{t_R/t_L}$
Hatano–Nelson eigenvector growth per site, $N = 12, 20, 30$	1.7321	$\sqrt{3}$
NH-SSH GBZ radius, $t_1 = 0.3 \ldots 2.0$	all 12 printed digits	$\sqrt{\lvert t_1-\gamma/2\rvert/\lvert t_1+\gamma/2\rvert}$
$\nu_{\text{non-Bloch}}$ vs zero-mode pairs ($N = 200$), 1001 points	equal except 37 points within 0.072 of a transition	equal
ν_{Bloch} at the same points	1/2 between Bloch closings; differs at 665 points	—
$\gamma = 0$	GBZ = unit circle, ν = SSH winding	same
Range-2 chain: max/min $\lvert\beta\rvert$ on GBZ	1.2786/1.0719 = 1.193	> 1 (not a circle)
Range-2, $N = 80$: distance of open spectrum to E(GBZ)	2.0×10^{-3} (to periodic loop: 0.44)	$\to 0$
Exponent of max root gap vs N	-0.980	-1
$(N+1)\ln\lvert\beta_3/\beta_2\rvert$ vs $\ln\lvert C(E)\rvert$	median $\le 10^{-12}$ for $N \ge 56$	equal
Circular GBZ in t_2 scan	only at $t_2 = 0.075$, max/min = 1.000000000000	$t_{-2}(t_1/t_{-1})^2 = 0.075$
`eig` vs 40-digit arithmetic, $N = 100$	plain 2.7×10^{-7}; gauged $\le 10^{-13}$	—

2 Theory and derivations

2.1 Toeplitz chains and the characteristic equation

We consider a single band on sites $n = 1, \ldots, N$ with translation-invariant hoppings of finite range,

$$H_{n,n+k} = t_k, \qquad -p \le k \le q, \tag{1}$$

so t_{-k} moves a particle k sites to the right. The open chain is a banded Toeplitz matrix. In the bulk the eigenvalue equation is the recursion

$$\sum_{k=-p}^{q} t_k\, \psi_{n+k} = E\, \psi_n. \tag{2}$$

Its solutions are $\psi_n = \beta^n$ with

$$E(\beta) = \sum_{k=-p}^{q} t_k \beta^k. \tag{3}$$

For a given E, multiplying $E(\beta) - E = 0$ by β^p gives a polynomial of degree $p + q$, with roots sorted as

$$|\beta_1(E)| \le |\beta_2(E)| \le \cdots \le |\beta_{p+q}(E)|. \tag{4}$$

Bloch's theorem is the statement that, with periodic boundaries, only $|\beta| = 1$ is allowed, $\beta = e^{ik}$. With open boundaries no such restriction exists: the eigenvector is a combination of all $p + q$ roots, and which roots dominate is decided by the boundaries.

Model	$E(\beta)$	p, q
Hatano–Nelson	$t_R\beta^{-1} + t_L\beta$	1,1
Range-2	$t_{-2}\beta^{-2} + t_{-1}\beta^{-1} + t_1\beta + t_2\beta^2$	2,2
NH-SSH in $\lambda = E^2$	$t_2\big(t_1 - \frac{\gamma}{2}\big)\beta^{-1} + \big(t_1^2 - \frac{\gamma^2}{4} + t_2^2\big) + t_2\big(t_1 + \frac{\gamma}{2}\big)\beta$	1,1

Chiral two-band chains as one band. For the non-reciprocal SSH chain the Bloch Hamiltonian in terms of β has off-diagonal entries

$$h_1(\beta) = t_1 + \frac{\gamma}{2} + \frac{t_2}{\beta}, \qquad h_2(\beta) = t_1 - \frac{\gamma}{2} + t_2\beta, \tag{5}$$

and $\det[H(\beta) - E] = E^2 - h_1 h_2$. The two-band problem is therefore a one-band problem in $\lambda = E^2 = h_1(\beta)h_2(\beta)$, whose Laurent polynomial is the third row of the table, and the same solver applies.

2.2 The GBZ condition

Boundary conditions. The open-chain eigenvector is $\psi_n = \sum_{j=1}^{p+q} c_j \beta_j^n$. Extend the recursion to the p fictitious sites $n = 1-p, \dots, 0$ on the left and the q sites $n = N+1, \dots, N+q$ on the right, and require ψ to vanish there. That gives $p+q$ linear conditions on the $p+q$ coefficients c_j, and E is an eigenvalue iff their determinant vanishes.

Why two roots must have equal modulus. The left conditions involve β_j^n with $n \le 0$, the right conditions β_j^{N+n} with $n \ge 1$. For large N, the right conditions are dominated by the q largest roots and the left by the p smallest. A non-trivial solution of both sets requires that the p-th and $(p+1)$-th roots compete on equal footing, otherwise the conditions decouple and only the trivial solution remains. In the limit $N \to \infty$ the open spectrum therefore lies on the set of E for which

$$|\beta_p(E)| = |\beta_{p+1}(E)|. \tag{6}$$

The GBZ is the curve traced by β_p and β_{p+1} on this set; it reduces to the unit circle for Hermitian chains. A precise statement is the Schmidt–Spitzer theorem [8].

The θ-scan. If β and $\beta e^{i\theta}$ both lie on the GBZ at the same energy, then $E(\beta) = E(\beta e^{i\theta})$. For one band this is a single polynomial equation in β:

$$\beta^p \big[E(\beta) - E(\beta e^{i\theta})\big] = \sum_{k=-p}^{q} t_k \big(1 - e^{ik\theta}\big)\beta^{k+p} = 0. \tag{7}$$

The $k = 0$ term drops out, and the polynomial has degree $p+q$. For each $\theta \in (0, 2\pi)$, every root β is a candidate. It is accepted iff, at $E = E(\beta)$, exactly $p-1$ of the $p+q$ characteristic roots are strictly smaller in modulus and $q-1$ strictly larger; the two remaining roots are then β and $\beta e^{i\theta}$, in positions p and $p+1$.

2.3 Hatano–Nelson chain

With $t_{-1} = t_R$ and $t_1 = t_L$, the characteristic equation is

$$t_L \beta^2 - E\beta + t_R = 0, \qquad \beta_1 \beta_2 = \frac{t_R}{t_L} \ \text{ for every } E. \tag{8}$$

Equal moduli then force $|\beta_1| = |\beta_2| = r$ with

$$r = \sqrt{|t_R / t_L|}. \tag{9}$$

The GBZ is a circle. To find the open spectrum exactly, substitute $\psi_n = r^n \phi_n$ in the bulk equation:

$$\frac{t_R}{r}\,\phi_{n-1} + t_L r\,\phi_{n+1} = E\,\phi_n, \qquad \frac{t_R}{r} = t_L r = \sqrt{t_R t_L}. \tag{10}$$

ϕ obeys a Hermitian chain with hopping $\sqrt{t_R t_L}$ and the same open boundaries, so

$$\psi_n \propto r^n \sin\frac{\pi m n}{N+1}, \qquad E_m = 2\sqrt{t_R t_L}\cos\frac{\pi m}{N+1}, \tag{11}$$

exactly at every N. For $t_R > t_L$, $r > 1$ and every eigenvector grows to the right. The periodic spectrum $E(e^{ik}) = (t_R + t_L)\cos k + i(t_L - t_R)\sin k$ is an ellipse, which the open chain never sees.

2.4 Non-reciprocal SSH chain: GBZ circle

The product of the two roots of the λ-quadratic is $(t_1 - \gamma/2)/(t_1 + \gamma/2)$, independent of λ. The same argument as for Hatano–Nelson gives a circular GBZ,

$$r = \sqrt{\frac{|t_1 - \gamma/2|}{|t_1 + \gamma/2|}} = \frac{1}{|s|}, \tag{12}$$

where s is the scaling factor of the nh-SSH similarity map. On the circle,

$$\lambda\big(re^{i\varphi}\big) = \big|v' + t_2 e^{i\varphi}\big|^2, \qquad v' = \sqrt{t_1^2 - \gamma^2/4}, \tag{13}$$

so $E(\mathrm{GBZ})$ is exactly the band of the Hermitian $\mathrm{SSH}(v', t_2)$ chain. The similarity map and the GBZ are the same statement, seen from real space and from β-space.

2.5 Non-Bloch winding number

The winding of the chiral chain is defined on a contour $|\beta| = \rho$ as

$$\nu = \frac{1}{2}\big(W[h_2] - W[h_1]\big), \qquad W[h] = \frac{1}{2\pi i}\oint_{|\beta|=\rho} \frac{h'(\beta)}{h(\beta)}\, d\beta. \tag{14}$$

The Bloch winding uses $\rho = 1$; the non-Bloch winding uses the GBZ, $\rho = r$. By the argument principle $W[h]$ is the number of zeros of h inside the contour minus the order of its poles there.

Zero counting. Write $a = t_1 + \gamma/2$ and $b = t_1 - \gamma/2$.

- $h_2 = b + t_2\beta$ has one zero, at $|\beta| = |b|/t_2$, and no pole.
- $h_1 = (a\beta + t_2)/\beta$ has one zero, at $|\beta| = t_2/|a|$, and a simple pole at $\beta = 0$.

Using $r^2 = |b|/|a|$, the positions of the zeros relative to the GBZ are

$$\frac{|b|}{t_2} < r \iff |ab| < t_2^2, \qquad \frac{t_2}{|a|} < r \iff |ab| > t_2^2. \tag{15}$$

If $|ab| < t_2^2$, the zero of h_2 is inside and that of h_1 outside, so $W[h_2] = 1$, $W[h_1] = 0 - 1 = -1$ and $\nu = 1$. Otherwise the zero of h_2 is outside and that of h_1 inside, $W[h_2] = 0$, $W[h_1] = 1 - 1 = 0$, and $\nu = 0$. Since $ab = t_1^2 - \gamma^2/4$,

$$\nu = 1 \iff \big|t_1^2 - \gamma^2/4\big| < t_2^2. \tag{16}$$

The absolute value matters. For $\gamma > 2t_2$ it produces the second transition at $t_1 = \sqrt{\gamma^2/4 - t_2^2}$; the form without absolute value misses it (at $\gamma = 3$, $t_1 = 0.9$ it predicts $\nu = 1$, but the chain has $\nu = 0$). At $\gamma = 0$, $r = 1$ and ν is the Hermitian SSH winding.

Bloch winding. On $|\beta| = 1$ the same counting gives

$$\nu_{\mathrm{Bloch}} = \tfrac{1}{2}\Big(\big[\,|b| < t_2\,\big] - \big[\,|a| > t_2\,\big] + 1\Big), \tag{17}$$

where $[\cdot]$ is 1 if the condition holds and 0 otherwise. It equals $1/2$ whenever exactly one zero lies on the "wrong" side of the unit circle; for $\gamma = 4/3$ this is the whole interval $1/3 < t_1 < 5/3$. The Bloch winding is then not even an integer.

2.6 Spectral winding and the GBZ

The GBZ can also be characterised by a winding number. For a reference energy E and a contour $|\beta| = \rho$ that avoids the roots, let

$$w_\rho(E) = \frac{1}{2\pi i} \oint_{|\beta|=\rho} \frac{E'(\beta)}{E(\beta) - E}\, d\beta. \tag{18}$$

$E(\beta) - E$ has a pole of order p at $\beta = 0$ and $p+q$ zeros, the characteristic roots. By the argument principle,

$$w_\rho(E) = \#\{j : |\beta_j(E)| < \rho\} - p. \tag{19}$$

- On the Brillouin zone, $\rho = 1$. For E inside the loop traced by the periodic spectrum, $w_1(E) \neq 0$: the number of roots inside the unit circle differs from p. This nonzero winding is the topological origin of the skin effect [5, 6], and its sign selects the end at which the eigenvectors accumulate.
- For any ρ strictly between $|\beta_p(E)|$ and $|\beta_{p+1}(E)|$, exactly p roots lie inside and $w_\rho(E) = 0$. The open-chain eigenvalues lie where this window closes, $|\beta_p| = |\beta_{p+1}|$, which is the GBZ condition. Moving the contour from the unit circle to the GBZ removes the spectral winding, which is why invariants computed on the GBZ, not on the unit circle, describe the open chain.

2.7 The Schmidt–Spitzer theorem

The statement used above has a precise mathematical form [8, 9]. Let T_N be the $N \times N$ banded Toeplitz matrix with symbol $E(\beta)$. Then, as $N \to \infty$, the eigenvalues of T_N accumulate on the set

$$\Lambda = \big\{E \in \mathbb{C} :\ |\beta_p(E)| = |\beta_{p+1}(E)|\big\}, \tag{20}$$

which is a finite union of analytic arcs. Λ is invariant under the imaginary gauge $\beta \to \rho\beta$, as it must be, since that gauge is a similarity transformation of every T_N. The theorem gives the limit set but not the rate of approach; Section 2.8 supplies that.

2.8 Leading finite-size term

The determinant condition of the open chain can be written out explicitly. Order the unknowns $c_1, \ldots, c_{p+q}$ by root modulus. The first p rows (left boundary) have entries β_j^n, $n = 1-p, \ldots, 0$; the last q rows (right boundary) have entries $\beta_j^{N+1}\beta_j^{n-1}$, $n = 1, \ldots, q$.

Laplace expansion. Expanding the determinant along the first p rows gives a sum over column subsets S of size p:

$$\det = \sum_S \pm T(S)\, B(S^c) \prod_{j \in S^c} \beta_j^{N+1}, \tag{21}$$

where $T(S)$ is the $p \times p$ minor of the top rows in columns S, and $B(S^c)$ the $q \times q$ minor of the bottom rows (without the factor β_j^{N+1}) in the complementary columns.

Dominant terms. For large N the product $\prod_{S^c} \beta_j^{N+1}$ is largest when S^c contains the q largest roots, i.e. $S_A = \{1, \ldots, p\}$. Near the GBZ, where $|\beta_p| \approx |\beta_{p+1}|$, the term with β_p and β_{p+1} exchanged, $S_B = \{1, \ldots, p-1, p+1\}$, is equally large. All other terms are smaller by at least e^{-Nd}, with

$$d = \min\Big(\ln\Big|\frac{\beta_p}{\beta_{p-1}}\Big|,\ \ln\Big|\frac{\beta_{p+2}}{\beta_{p+1}}\Big|\Big). \tag{22}$$

Result. Setting the sum of the two dominant terms to zero and dividing,

$$\Big(\frac{\beta_{p+1}}{\beta_p}\Big)^{N+1} = -\frac{T(S_B)\,B(S_B^c)}{T(S_A)\,B(S_A^c)} \equiv C(E). \tag{23}$$

Taking the modulus and the logarithm,

$$(N+1)\ln\Big|\frac{\beta_{p+1}(E)}{\beta_p(E)}\Big| = \ln|C(E)|. \tag{24}$$

The left side is $(N+1)$ times the root gap, which vanishes on $E(\mathrm{GBZ})$. Each finite-N eigenvalue therefore misses the continuum condition by exactly $\ln|C(E)|/(N+1)$, with C computed from the characteristic roots alone, and with corrections $O(e^{-Nd})$. The phase of the same equation quantises the eigenvalues along the arcs, the analogue of $k = \pi m/(N+1)$.

Two special cases follow. For $p = q = 1$ both minors are 1×1 and $C = 1$ exactly, so Hatano–Nelson eigenvalues sit on $E(\mathrm{GBZ})$ at every N, as found above. At junctions where three arcs of $E(\mathrm{GBZ})$ meet, a third root has the same modulus, $d \to 0$, and a three-root expansion would be needed.

2.9 When is the GBZ a circle?

Imaginary gauge. The substitution $\psi_n = \rho^n \phi_n$ is a similarity transformation by $D = \mathrm{diag}(\rho^n)$. It maps $t_k \to t_k \rho^k$, i.e. $E(\beta) \to E(\rho\beta)$, so the GBZ is rescaled to GBZ/ρ while the spectrum is unchanged.

Criterion. Suppose some ρ makes the chain reciprocal, $t_k\rho^k = t_{-k}\rho^{-k}$ for all k. Then the gauged Laurent polynomial is symmetric under $x \to 1/x$, its roots come in pairs $(x, 1/x)$, and $|x_p| = |x_{p+1}|$ forces $|x| = 1$. The GBZ is then the circle $|\beta| = \rho$. For the range-2 chain the conditions for $k = 1$ and $k = 2$ are compatible iff

$$t_2 = t_{-2}\Big(\frac{t_1}{t_{-1}}\Big)^2, \qquad \rho = \sqrt{t_{-1}/t_1}. \tag{25}$$

For $p = q = 1$ a (possibly complex) ρ always exists, which is why Hatano–Nelson and the chiral SSH chain always have circular GBZs. The GBZ is the *unit* circle iff the chain is already reciprocal; equivalently, the periodic spectrum has zero spectral winding about every E, the no-skin-effect criterion [5, 6].

2.10 Why the gauge improves the eigensolver

Eigenvalue errors scale with the non-normality of the matrix, which for a Toeplitz chain is set by how strongly the eigenvectors grow across the chain. Without a gauge the range-2 eigenvectors grow as $|\beta|^n$ with $|\beta| \in [1.07, 1.28]$. With

$$\rho = \sqrt{\min|\beta|\ \max|\beta|} = 1.171 \tag{26}$$

the per-site growth is only $|\beta|/\rho \in [0.92, 1.09]$; only the non-circular part of the GBZ remains. For a circular GBZ the gauge $\rho = r$ removes all non-normality: it is the nh-SSH similarity map seen from the GBZ side.

3 Numerical methods

3.1 GBZ by the θ-scan

1. The Laurent coefficients are trimmed of zero end terms, so that p and q are the true ranges.

2. The scan angles are $\theta = \pi(1 - \cos u)$ with u uniform on $(0, \pi)$, which clusters points near $\theta \to 0$ and 2π, where the GBZ has its branch points; 4000 angles are used.
3. For each θ the polynomial of Eq. (7) is solved by the eigenvalues of its companion matrix.
4. For each root, $E = E(\beta)$ is evaluated and the full characteristic polynomial at that E is solved. The root is accepted if exactly $p - 1$ roots are smaller and $q - 1$ larger in modulus, with a relative tolerance of 10^{-9}.

Counting, rather than sorting the roots and taking indices p and $p + 1$, is robust when two roots are nearly degenerate, since a sort would then pick an arbitrary order. The accepted points, mapped through $E(\beta)$, give $E(\mathrm{GBZ})$.

Accuracy of the roots. Companion-matrix eigenvalues are the exact roots of a polynomial whose coefficients are perturbed at the level of rounding. A simple root is then accurate to $O(\epsilon_{\mathrm{mach}})$, but a double root only to $O(\epsilon_{\mathrm{mach}}^{1/2}) \approx 10^{-8}$. At $\theta \to 0$ and 2π the pair β, $\beta e^{i\theta}$ merges into a double root, so the scan grid excludes the endpoints, and the acceptance tolerance 10^{-9} is applied to the *other* roots, which stay simple. The root $\beta = 0$, which appears when the lowest coefficient vanishes, is discarded. The accepted points are sorted by $\arg\beta$ to trace the curve.

Distance to $E(\mathrm{GBZ})$. $E(\mathrm{GBZ})$ is stored as a densely sampled polyline. The distance of an eigenvalue from it is the minimum, over all segments, of the distance to the nearest point on the segment, obtained by projecting onto the segment and clamping the projection parameter to $[0, 1]$.

3.2 Winding numbers and edge states

The winding is computed as a wrapped phase sum on 2001 points of the contour $|\beta| = \rho$. Because a zero of h_i on the contour makes the winding undefined, the zeros of h_1 and h_2 are located exactly and the routine returns NaN if one lies on the contour, instead of relying on sampling. An independent value is obtained by counting zeros and poles inside the contour. The edge-state count is the number of $\pm E$ pairs with $|E| < 10^{-8}$ in the exact nh-SSH open spectrum at $N = 200$, obtained from the Hermitian partner chain.

3.3 Finite-size analysis

For each open-chain eigenvalue E the characteristic roots are recomputed, giving the root gap $\ln|\beta_{p+1}/\beta_p|$, the separation d from the nearest third root, and $\ln|C(E)|$ from the $p \times p$ and $q \times q$ minors defined above. The distance of each eigenvalue from $E(\mathrm{GBZ})$ is the minimum distance to a densely sampled curve. The exponent of the convergence is a linear fit of log(max gap) against $\log N$.

3.4 Gauged diagonalisation and reference values

The open-chain matrix is built with a gauge, $t_k \to t_k \rho^k$, and diagonalised with the general dense eigensolver. Two independent checks are used: the eigenvalues are compared with those for a second gauge 1.03ρ at every N (both are exact similarity transforms, so any difference is numerical error), and with 40-digit reference eigenvalues for $N = 40$ to 100.

4 Verification

The tests check each analytic statement of Section 2 against an independent numerical computation.

1. The model. Zero end coefficients must be trimmed; the Hatano–Nelson matrix elements must be correct; the bulk rows of the open chain must reproduce the Laurent symbol of Eq. (3); and the gauge must be an exact similarity transform (identical spectrum). The λ-symbol of the SSH chain must equal $h_1 h_2$ and match the Bloch energies of the nh-SSH project.

2. Circular GBZs. The θ-scan must reproduce $\sqrt{t_R/t_L}$ and $1/|s|$. The Hatano–Nelson open spectrum must match its closed form, the eigenvectors must be r^n times sines, and the eigenvalues must lie on $E(\mathrm{GBZ})$ but not on $E(|\beta| = 1)$. For the SSH chain an independent check runs in the opposite direction: for each bulk open eigenvalue, the characteristic roots must satisfy $|\beta_1| = |\beta_2| = r$. The zero modes, as a contrast, must lie off the continuum.

3. Non-circular GBZ. The range-2 GBZ must have max/min $|\beta| > 1$; it must transform covariantly under the gauge (GBZ→GBZ/ρ); a reciprocal chain must give the unit circle; and a range-2 chain tuned to satisfy the circular criterion must give an exact circle of radius $\sqrt{2}$, while the generic chain is not a circle.

4. Winding. The phase sum must match zero counting; the winding of a monomial β^m must be m; the Hermitian limit must reduce to the SSH winding; ν must match the closed form Eq. (16) and count the edge states of the open chain; it must be undefined exactly at the transitions; and the form without absolute value must be shown to miss the second transition.

5. Finite size. The open spectrum of the range-2 chain must converge to $E(\mathrm{GBZ})$ as $1/N$. Eq. (24) must hold for each eigenvalue within $2e^{-Nd}$, and $\ln|C|$ must vanish identically for Hatano–Nelson.

6. Accuracy. The gauged eigenvalues must match the 40-digit references, and the plain eigensolver must be shown to drift at $N = 100$, so that the improvement is measured rather than assumed. The figure script must give bitwise-identical results on two runs.

5 Results

5.1 Hatano–Nelson

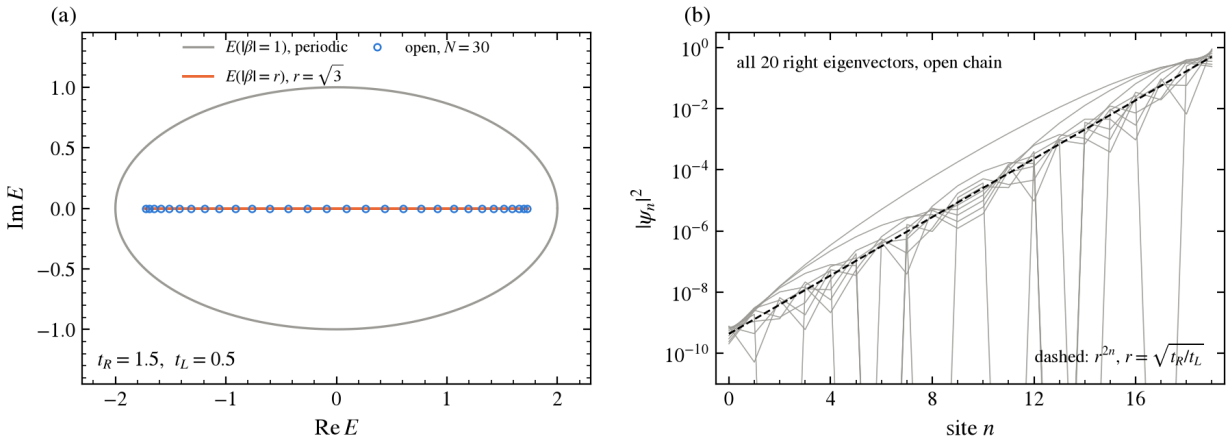


Figure 1: $t_R = 1.5$, $t_L = 0.5$. (a) Periodic spectrum (grey ellipse); E on the GBZ $|\beta| = \sqrt{3}$ collapses to a segment (orange) on which the 30 open eigenvalues lie. (b) All 20 right eigenvectors of a 20-site chain grow to the right as r^{2n} (dashed).

The scan gives $\sqrt{t_R/t_L}$ to 2×10^{-15} for three parameter sets. Even this simplest chain is non-normal: the plain eigensolver at $N = 30$ errs by 7×10^{-11} because $r^N = 3^{15}$.

5.2 Non-reciprocal SSH chain

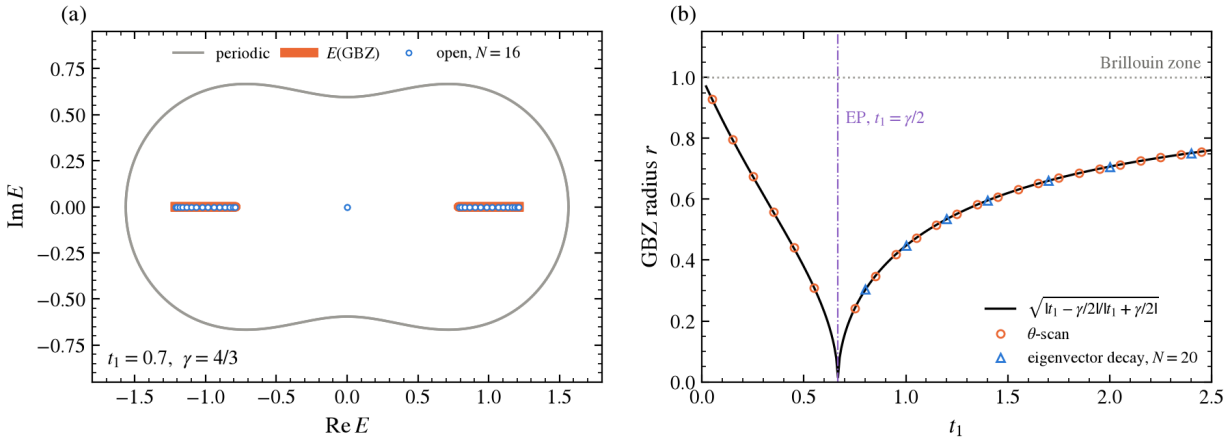


Figure 2: (a) $t_1 = 0.7$, $\gamma = 4/3$: periodic loop (grey), $E = \pm\sqrt{\lambda}$ on the GBZ (orange, two real segments), exact open spectrum $N = 16$ (circles) plus the zero-mode pair. (b) GBZ radius vs t_1: formula (line), θ-scan (circles), bulk eigenvector decay from nh-SSH (triangles). At $t_1 = \gamma/2$ the GBZ shrinks to $\beta = 0$.

The θ-scan reproduces the radius in all 12 printed digits at $t_1 = 0.3, 0.7, 1.0, 1.4, 2.0$, including $t_1 < \gamma/2$ where v' is imaginary. The zero modes are *not* on the continuum: their root gap is finite.

5.3 Bloch versus non-Bloch winding

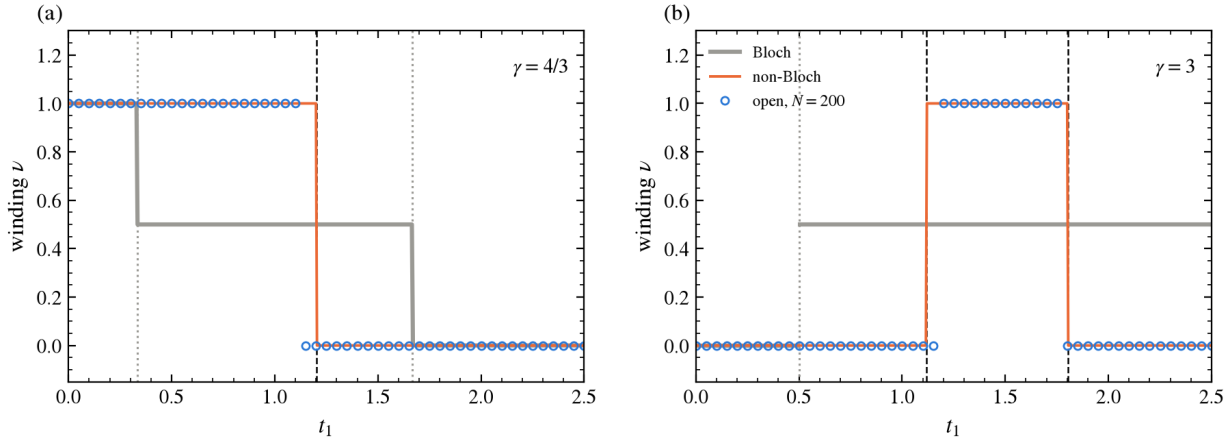


Figure 3: Winding vs t_1: Bloch (grey), non-Bloch (orange), zero-mode pairs at $N = 200$ (circles); dashed: OBC transitions; dotted: Bloch closings. (a) $\gamma = 4/3$. (b) $\gamma = 3$.

(t_1, γ)	ν_{Bloch}	$\nu_{\text{non-Bloch}}$	zero-mode pairs	$\min \lvert E \rvert$
(0.2, 4/3)	1	1	1	3×10^{-17}
(0.7, 4/3)	1/2	1	1	3×10^{-17}
(1.0, 4/3)	1/2	1	1	3×10^{-16}
(1.3, 4/3)	1/2	0	0	0.12
(2.0, 4/3)	0	0	0	0.89
(0.9, 3)	1/2	0	0	0.67
(1.4, 3)	1/2	1	1	4×10^{-16}
(2.2, 3)	1/2	0	0	0.61

The only disagreements (13 and 24 points) lie within 0.062 and 0.072 of a transition, where the $N = 200$ splitting has not yet dropped below 10^{-8}. At $t_1 = 1$ the zeros are at $|\beta| = 0.333$ (h_2) and 0.600 (h_1); the unit circle encloses both ($\nu = 1/2$), while the GBZ, $r = 0.447$, separates them ($\nu = 1$).

5.4 A non-circular GBZ

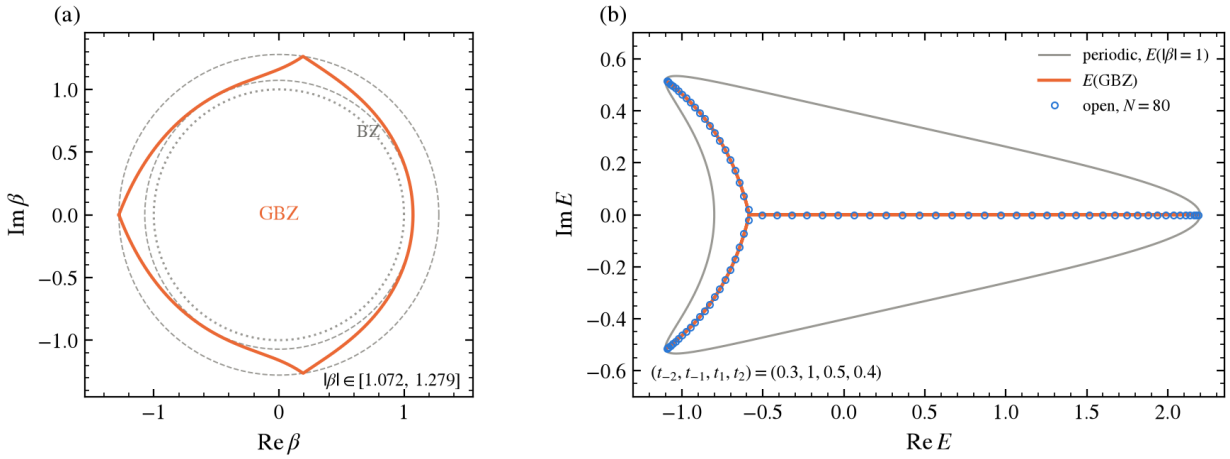


Figure 4: $(t_{-2}, t_{-1}, t_1, t_2) = (0.3, 1, 0.5, 0.4)$. (a) GBZ (orange) between $|\beta| = 1.0719$ and 1.2786; dotted: BZ. (b) Periodic spectrum (grey), E(GBZ) (orange), open chain $N = 80$ (blue, gauged).

The GBZ has three cusps at its largest $|\beta|$, where $E = -0.583$ and the root moduli are

$$|\beta_{1..4}| = (0.3585,\ 1.2786,\ 1.2786,\ 1.2794),$$

a third root within 0.1%. The cusps map to the junction where three arcs of E(GBZ) meet.

5.5 How fast the open spectrum reaches $E(\mathrm{GBZ})$

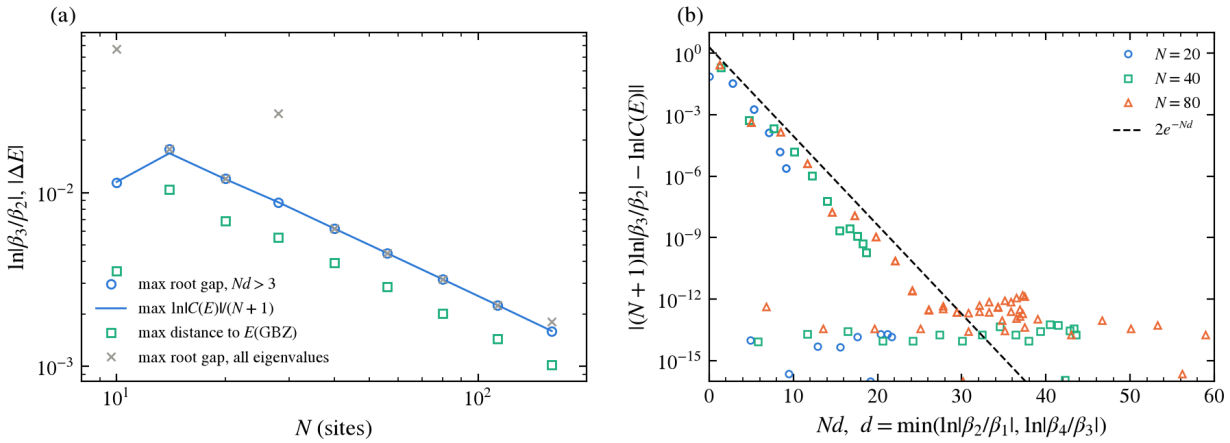


Figure 5: (a) Largest root gap of open eigenvalues ($Nd > 3$, circles), prediction $\max \ln|C|/(N+1)$ (line), largest E-plane distance (squares), largest gap over all eigenvalues (crosses). (b) $|(N+1)\cdot\mathrm{gap} - \ln|C(E)||$ per eigenvalue vs Nd at $N = 20, 40, 80$; dashed: $2e^{-Nd}$.

N	eigenvalues with $Nd > 3$	max gap	$\max \ln|C|/(N+1)$	max E-distance
20	17/20	1.203×10^{-2}	1.194×10^{-2}	6.9×10^{-3}
40	38/40	6.210×10^{-3}	6.210×10^{-3}	3.9×10^{-3}
80	78/80	3.161×10^{-3}	3.161×10^{-3}	2.0×10^{-3}
160	158/160	1.595×10^{-3}	1.595×10^{-3}	1.0×10^{-3}

The finite-size error can be stated exactly: every eigenvalue misses the GBZ condition by $\ln|C(E)|/(N+1)$, and the residual of Eq. (24) falls as e^{-Nd} until it hits round-off at 10^{-13}. The fitted exponent is -0.980 rather than -1 because eigenvalues creep towards the junction where $|C|$ is largest. Including the one or two eigenvalues next to the junction ($d \to 0$), the exponents drop to -0.910 (gap) and -0.760 (E-distance); a three-root version of Eq. (24) would be needed there.

5.6 Deforming the GBZ

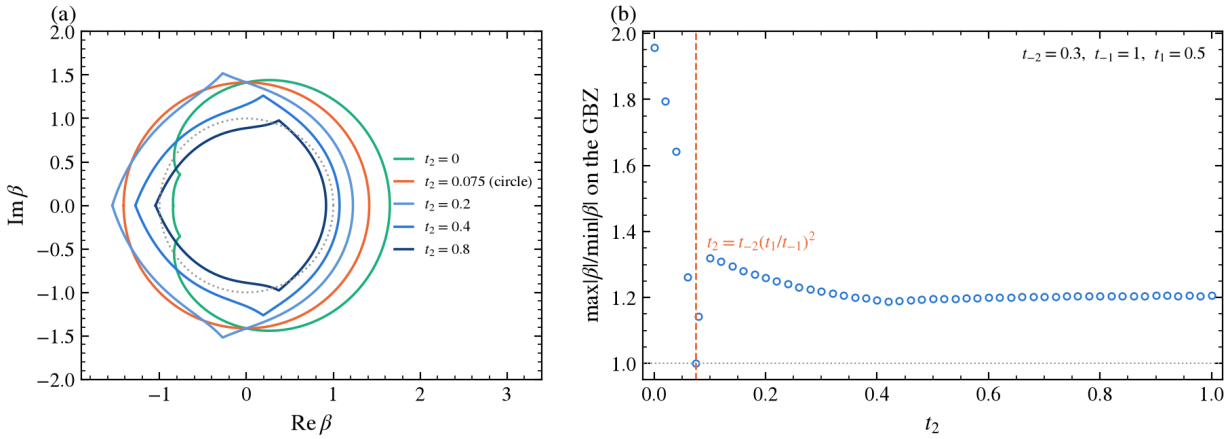


Figure 6: $t_{-2} = 0.3$, $t_{-1} = 1$, $t_1 = 0.5$, varying t_2. (a) GBZ for five t_2. (b) max/min $|\beta|$ from 52 scans; dashed: predicted circular point.

t_2	0	0.075	0.2	0.4	0.8	1.0
max/min $\lvert\beta\rvert$	1.958	1.000	1.259	1.193	1.205	1.206

The circle appears exactly at the predicted $t_2 = 0.075$ with radius $\sqrt{2}$, and nowhere else.

5.7 Numerical accuracy

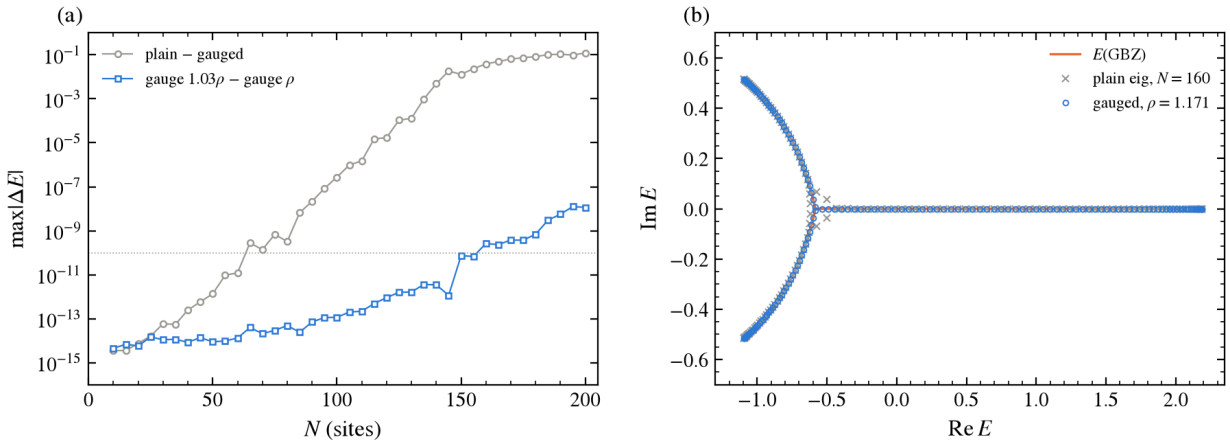


Figure 7: (a) Plain vs gauged eigenvalues (grey), gauge 1.03ρ vs ρ (blue), both vs 40-digit references (stars). (b) $N = 160$: plain eigenvalues scatter off $E(\mathrm{GBZ})$ near the junction; gauged eigenvalues stay on it.

N	plain vs 40-digit	gauged vs 40-digit	gauge vs gauge
40	2.5×10^{-13}	1.0×10^{-14}	8.4×10^{-15}
60	1.2×10^{-11}	8.2×10^{-15}	1.3×10^{-14}
80	3.2×10^{-10}	1.6×10^{-14}	4.9×10^{-14}
100	2.7×10^{-7}	7.5×10^{-14}	1.2×10^{-13}
160	—	—	2.6×10^{-10}

The plain eigensolver passes 10^{-10} at $N = 65$ and reaches 0.04 by $N = 160$. Unlike the nh-SSH chain, there is no step at the 75-site LAPACK crossover: the error grows smoothly. The remaining gauged error is set by the non-circular part of the GBZ, which no diagonal gauge can remove.

6 Discussion

What breaks on the unit circle. The spectrum ($E(|\beta| = 1)$ encloses area; the open chain lives on arcs inside it), the eigenvectors (every bulk state is β^n with $|\beta| \neq 1$) and the topology (Bloch winding 1/2, changing at Bloch closings where nothing happens to the open chain).

BZ, GBZ and similarity. $D^{-1}HD$ has the same eigenvalues but eigenvectors $D^{-1}\psi$; $D = \mathrm{diag}(\rho^n)$ rescales the GBZ. A circular GBZ can be mapped onto the BZ, turning the problem Hermitian (nh-SSH project); a non-circular one cannot, which is also why a gauge improves the eigensolver but cannot fix it at arbitrary N.

Limitations and outlook. (i) Junction convergence needs a three-root boundary determinant. (ii) Generic two-band chains without chiral symmetry need the full resultant of two polynomials;

auxiliary-GBZ methods [7] are a natural comparison. (iii) With disorder there is no GBZ; whether $\langle \ln |t_R/t_L| \rangle$ still predicts skin accumulation and edge-state loss is open. (iv) For driven-dissipative cavity arrays the response is set by the Green's function [10]; how the two windings appear in directional amplification is a natural next project.

7 Conclusion

One θ-scan solver produces the GBZ for single-band, chiral two-band and longer-range chains, reproducing every analytic circle to 10^{-15}. The non-Bloch winding counts open-chain edge states where the Bloch winding gives a non-integer. A Laplace expansion of the boundary determinant yields the leading finite-size term with no free parameter, confirmed per eigenvalue to round-off. The exact circular-GBZ criterion unifies the Hatano–Nelson, nh-SSH and Hermitian cases and explains why an imaginary gauge restores eigensolver accuracy.

References

[1] S. Yao, Z. Wang, *Edge states and topological invariants of non-Hermitian systems*, Phys. Rev. Lett. **121**, 086803 (2018).

[2] K. Yokomizo, S. Murakami, *Non-Bloch band theory of non-Hermitian systems*, Phys. Rev. Lett. **123**, 066404 (2019).

[3] E. J. Bergholtz, J. C. Budich, F. K. Kunst, *Exceptional topology of non-Hermitian systems*, Rev. Mod. Phys. **93**, 015005 (2021).

[4] N. Hatano, D. R. Nelson, *Localization transitions in non-Hermitian quantum mechanics*, Phys. Rev. Lett. **77**, 570 (1996).

[5] N. Okuma, K. Kawabata, K. Shiozaki, M. Sato, *Topological origin of non-Hermitian skin effects*, Phys. Rev. Lett. **124**, 086801 (2020).

[6] K. Zhang, Z. Yang, C. Fang, *Correspondence between winding numbers and skin modes in non-Hermitian systems*, Phys. Rev. Lett. **125**, 126402 (2020).

[7] Z. Yang, K. Zhang, C. Fang, J. Hu, *Non-Hermitian bulk-boundary correspondence and auxiliary generalized Brillouin zone theory*, Phys. Rev. Lett. **125**, 226402 (2020).

[8] P. Schmidt, F. Spitzer, *The Toeplitz matrices of an arbitrary Laurent polynomial*, Math. Scand. **8**, 15 (1960).

[9] A. Böttcher, S. M. Grudsky, *Spectral Properties of Banded Toeplitz Matrices*, SIAM (2005).

[10] C. C. Wanjura, M. Brunelli, A. Nunnenkamp, *Topological framework for directional amplification in driven-dissipative cavity arrays*, Nat. Commun. **11**, 3149 (2020).